Notes on Computational Fluid Dynamics: General Principles

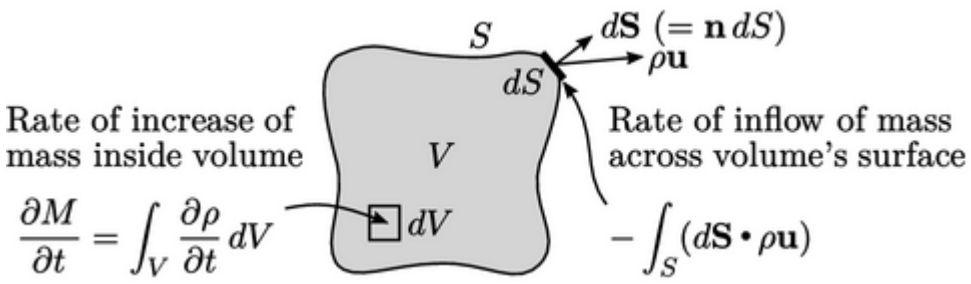
2.4 Conservation of mass

The law of conservation of mass can be written

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0} \tag{2.8}$$

where ρ is the mass density of the fluid. The equation can be derived by considering a volume V *fixed in space* (note !!), bounded by a surface S. If the volume is filled by a fluid with density ρ, its mass is $M = \int_V \rho \, dV$.

The rate of increase of mass inside the volume must match the rate of inflow of mass across the volume's surface. The latter is calculated by integrating the mass flux $\rho\phi$ over the surface, noting the negative sign due to $d\mathbf{S}$ pointing out of the volume.



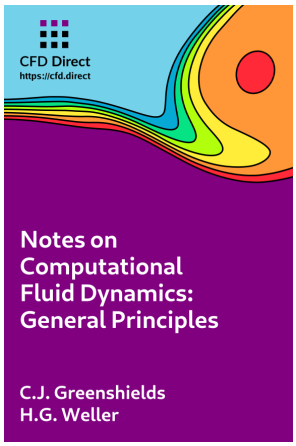
Gauss's Theorem relates surface and volume integrals by

$$\int_S (d\mathbf{S} \cdot \rho \mathbf{u}) = \int_V \nabla \cdot (\rho \mathbf{u}) \, dV. \tag{2.9}$$

Equating the rate of mass increase to rate of inflow and applying Eq. (2.9) gives

$$\int_V \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) \right] dV = 0. \tag{2.10}$$

Since the integral is valid for any volume V, it follows that the integrand (in $[\ldots]$) must equal 0, resulting in Eq. (2.8).



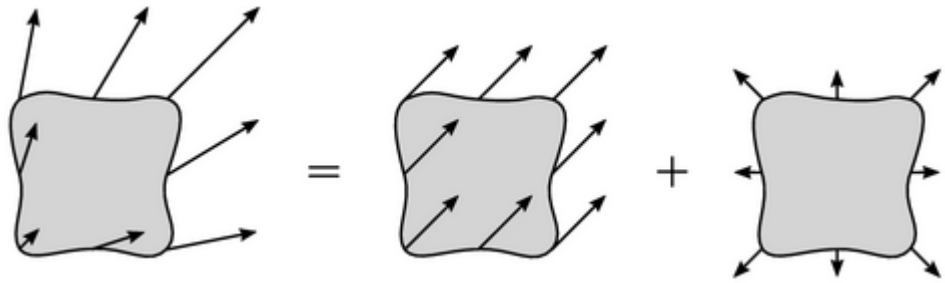
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Divergence

Divergence, denoted by $\nabla \bullet (\ldots)$, indicates the tendency of a vector field to point outward of a closed surface. For example, When the divergence of velocity $\nabla \bullet \mathbf{u}$ is positive, the fluid is expanding; negative divergence indicates contraction. Imagine a volume of fluid with a complex distribution of $\mathbf{u}$ at its bounding surface, below left. The $\nabla \bullet \mathbf{u}$ calculation will isolate the diverging component (right) from uniform flow (centre).



The divergence of velocity is calculated by integrating — *i.e.* summing — fluxes over the closed surface. To define divergence *at a point*, we consider the limiting case where the volume tends to zero. For a surface S that encloses a volume ΔV, divergence is the flux across the surface per unit volume, as $\Delta V \to 0$, *e.g.*

$$\nabla \bullet \mathbf{u} = \lim_{\Delta V \to 0} \frac{1}{\Delta V} \int_S (d\mathbf{S} \bullet \mathbf{u}). \tag{2.11}$$

The nabla operator

The nabla symbol ∇ can be considered a *vector operator*

$$\nabla \equiv \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right). \tag{2.12}$$

Divergence is the inner product with ∇, *e.g.*

$$\nabla \bullet \mathbf{u} = \frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z}. \tag{2.13}$$

Chris Greenshields Henry Weller 11th April 2022 Notes on CFD

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